Andrei Dumitrescu

Software Engineer | Java · Spring Boot · TypeScript · Node.js · REST APIs · PostgreSQL

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SUMMARY

Software engineer building backend web services and full-stack products in Java/Spring Boot and TypeScript/Node.js, with React and PostgreSQL. Technical lead on a production AI product: own the Node.js backend and REST APIs, the data and job-queue layer around the LLM service, payments and releases, leading 2 engineers. Previously 3 years of enterprise Java backend, REST/SOAP integration and CI/CD at Ness Digital Engineering, in Scrum/Agile teams (fraud/AML, logistics, tourism).

EXPERIENCE

Technical Lead — ZettaSource July 2025 – September 2026

Product studio — flagship AI app delivered for a client, plus internal prototypes. Engagement concluded September 2026; available immediately.

- Own end-to-end delivery of an AI companion app — TypeScript/Node.js (Deno) backend on Supabase/PostgreSQL, OpenAI-based AI features, RevenueCat payments, PostHog analytics, React/Vite/Capacitor client and store releases.
- Led 2 engineers through delivery: broke the architecture into tasks, set technical direction, held quality through code review, and turned the product owner's business logic into shipped features.
- Designed and built the backend — 80+ TypeScript edge functions (Node.js/Deno runtime), REST endpoints, row-level security, Auth with Apple/Google/Meta SSO, Storage with signed URLs, pg_cron jobs, WebSocket streaming chat, query-level caching and indexing — serving 2,800+ registered users.
- Designed the data model, user-context memory handling and retry-safe background processing around the OpenAI-based conversation service (15,000+ conversations, 95,000+ messages in production), plus the asynchronous job-queue contract so long-running media generation (ElevenLabs, Whisper, image/video) never blocks the app.
- Implemented monetization end-to-end — subscriptions, paywalls and an in-app coin economy — with RevenueCat and auditable transaction logic, driving ~3.5% free-to-paid conversion.
- Set up quality and release engineering: 470+ Vitest test files across client and edge functions, Maestro E2E flows, and a PR pipeline on GitHub Actions (lint, tsc, sharded tests, deno check, migration-sync check, iOS validation, Android build) with automatic edge-function deploys on merge — ~10 reverts across 450+ PRs.
- Built operational safety without a dedicated SRE: feature-flag kill switches, OTA live updates (Capawesome) and revert-driven rollback; scheduled health checks on AI-service failure rates, DB security/performance advisors and RevenueCat revenue and refunds; fixed production issues such as duplicate chat replies (unique index + conflict handler) and a stuck transaction blocking Realtime.
- Built the growth and compliance layer: server-side PostHog events and feature flags, OneSignal push (APNs/FCM), deterministic A/B assignment, GDPR deletion and audit trails.
- Built the React/Vite/Tailwind client with Capacitor (one codebase, both stores) and owned App Store Connect, TestFlight and Google Play releases.
- Prototyped Rust-based payment orchestration for the studio, forked from open-source Hyperswitch — PSP/gateway connectors, 3DS flow, authorization/capture, refunds, idempotent retries, PCI-aligned access controls.

Software Engineer (Freelance, part-time) — ZettaSource June 2024 – July 2025

Evenings & weekends, alongside a full-time role

- Built and maintained 8 e-commerce storefronts (Shopify, WooCommerce) for 3 clients — performance and structural work significantly improved search visibility and conversion; automated inventory and price monitoring, saving ~10 hours/week.

Software Engineer (Backend Java) — Ness Digital Engineering July 2022 – July 2025

Promoted twice: Trainee → Junior → Mid-level; consecutive client engagements in Scrum/Agile delivery teams

- Mid-level (Feb 2024 – Jul 2025): Backend Java developer on a fraud & anti-money-laundering solution on an IBM BAW-type BPM platform — BPMN 2.0 processes, REST/SOAP APIs and business rules (Java, Spring Boot, Scala, MSSQL, Oracle, Swagger/Postman), until the engagement concluded. Earlier in the period, refactored legacy Java/Selenium/Cucumber suites for another client, cutting CI regression time by 20%+.
- Junior (Mar 2023 – Feb 2024): Backend Java developer on a microservices-based maritime logistics platform (Spring Boot, Hibernate/JPA, Kafka, Maven/Gradle) tracking freight-exchange bids and vessel positions. Earlier in the period, built CI/CD for another client: Jenkins/Groovy pipelines with jgitver versioning across microservices on Docker/Kubernetes with SonarQube and Nexus.
- Trainee (Jul 2022 – Mar 2023): Full-stack development of a tourism web application for Timișoara with Traian Vuia International Airport (Java, MongoDB, JUnit 5, Mockito, Rest Assured, Swagger, Azure DevOps).

SKILLS

Main: Java, Spring Boot, TypeScript, Node.js, Deno, React, PostgreSQL, REST APIs, Docker, Kubernetes, AWS, Git, GitHub Actions, Jenkins, Scrum/Agile, LLM integration (OpenAI), AI-assisted development (Claude Code, Cursor)

Additional: Kafka, Hibernate/JPA, WebSockets, MSSQL, Oracle, MongoDB, Supabase, Next.js, Tailwind, Capacitor (iOS/Android), Scala, Rust, Python (FastAPI), Azure DevOps, Linux, Grafana, JUnit/Mockito, Selenium, Cucumber, BPM/BPMN 2.0 (IBM BAW-type), RevenueCat, PostHog, OneSignal, ElevenLabs, Whisper

EDUCATION

B.Sc. Computer Science — West University of Timișoara, 2020–2023 · Security labs: PortSwigger Web Security Academy, TryHackMe, Hack The Box